

HVAC System Optimization Report

Ottawa Commercial Building Case Study

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Executive Summary

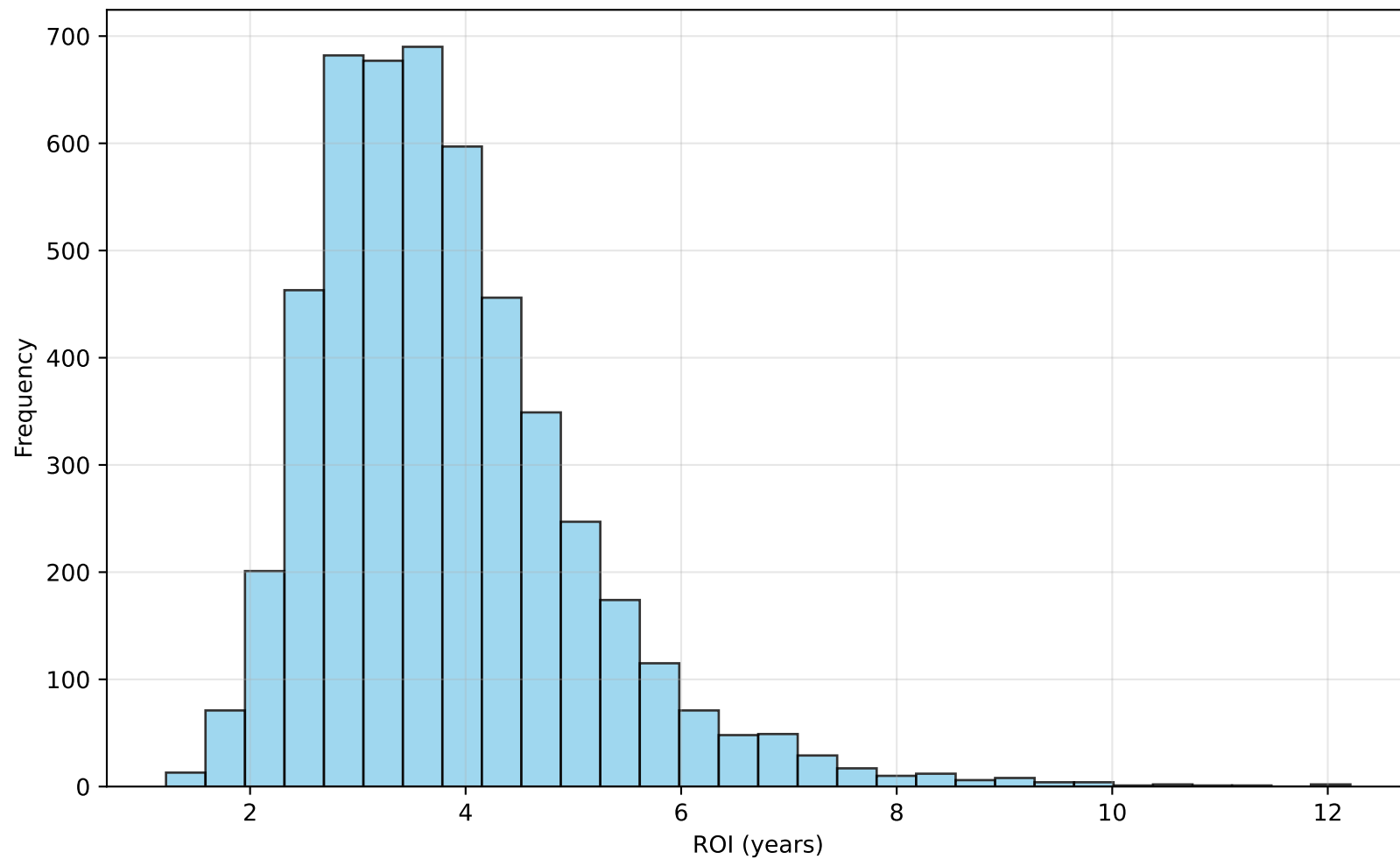
Baseline Annual Energy : 636,413 kWh
Baseline Annual Cost : \$76,369.55

Best Scenario: Combined Measures

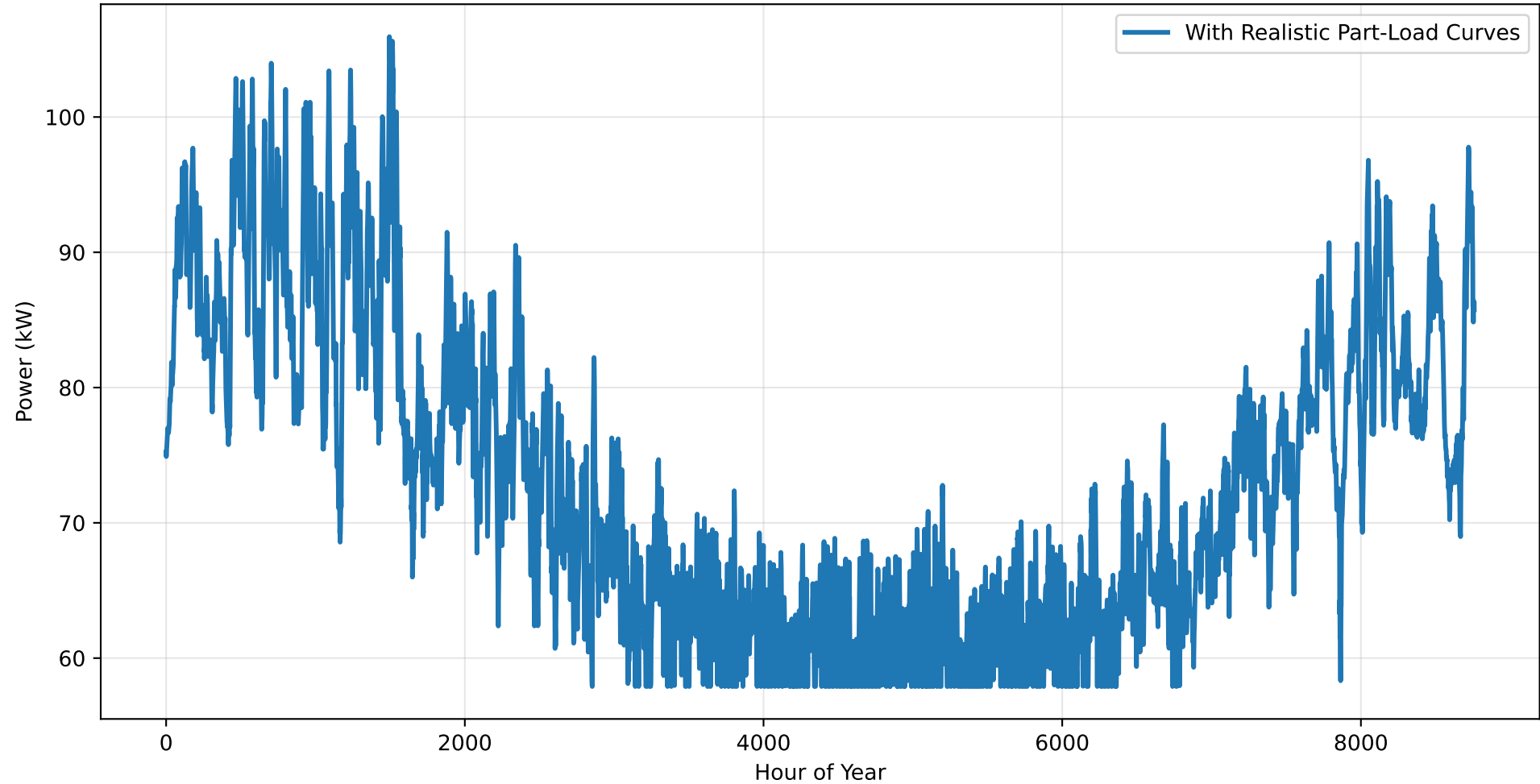
- Energy Reduction : 85,410 kWh (13.4%)
- Cost Savings : \$10,249.00
- Payback Period : 4.9 years
- CO₂ Reduction : 42.7 tonnes

Parametric Optimization found even better configuration saving an extra ~\$9,840/year.

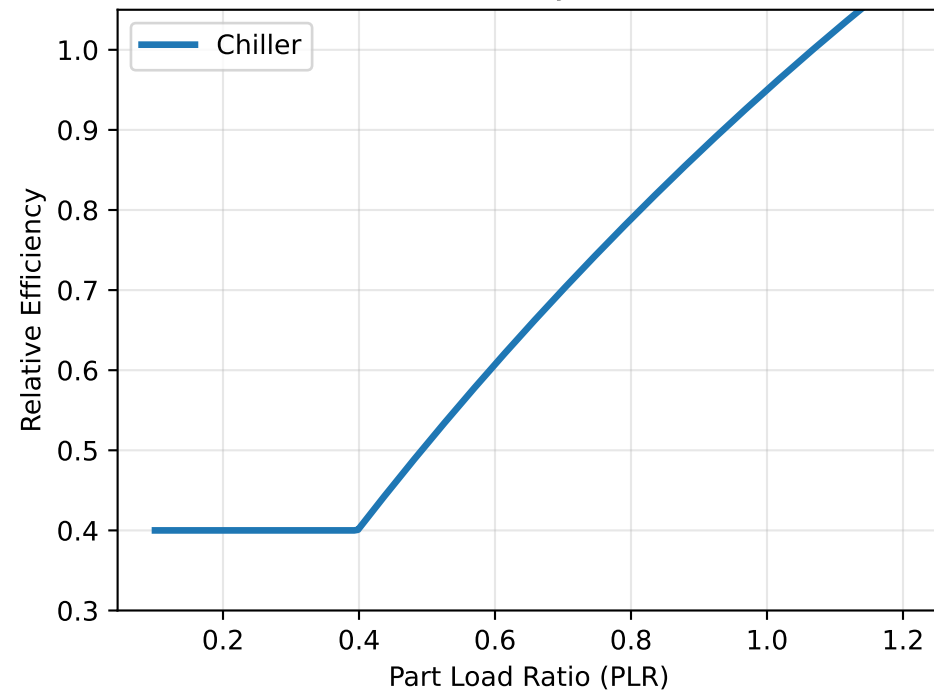
Monte Carlo Simulation - ROI Distribution (5000 runs)



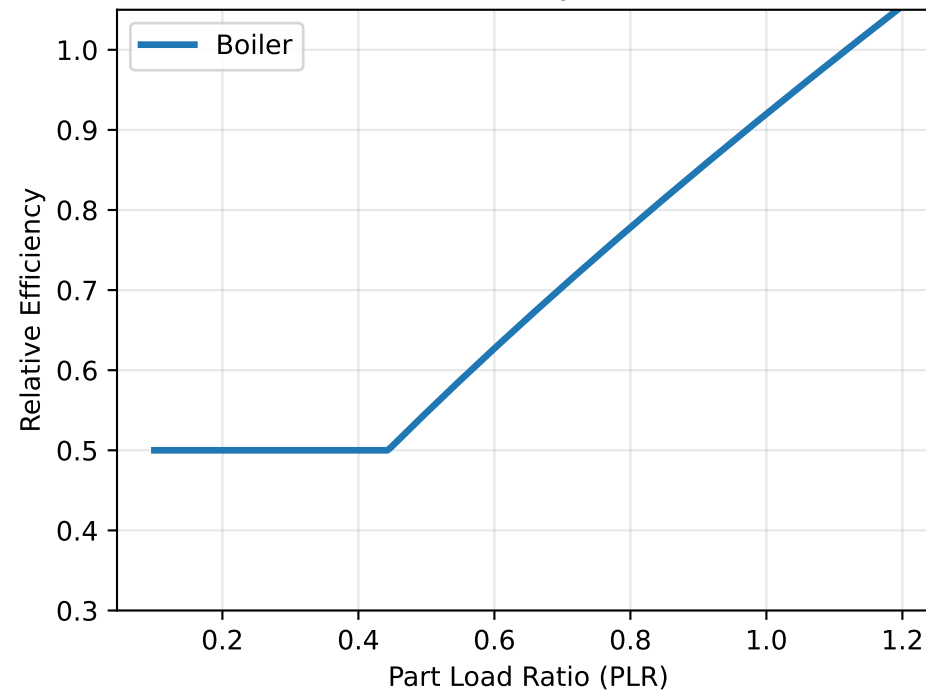
Impact of Part-Load Curves on Baseline Energy



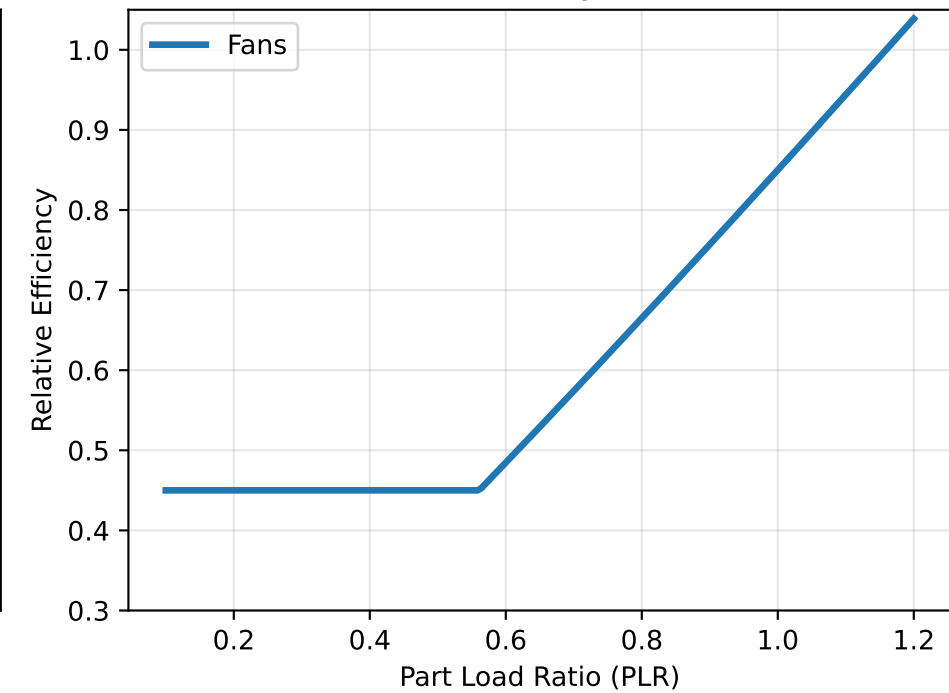
Part-Load Efficiency Curve: Chiller



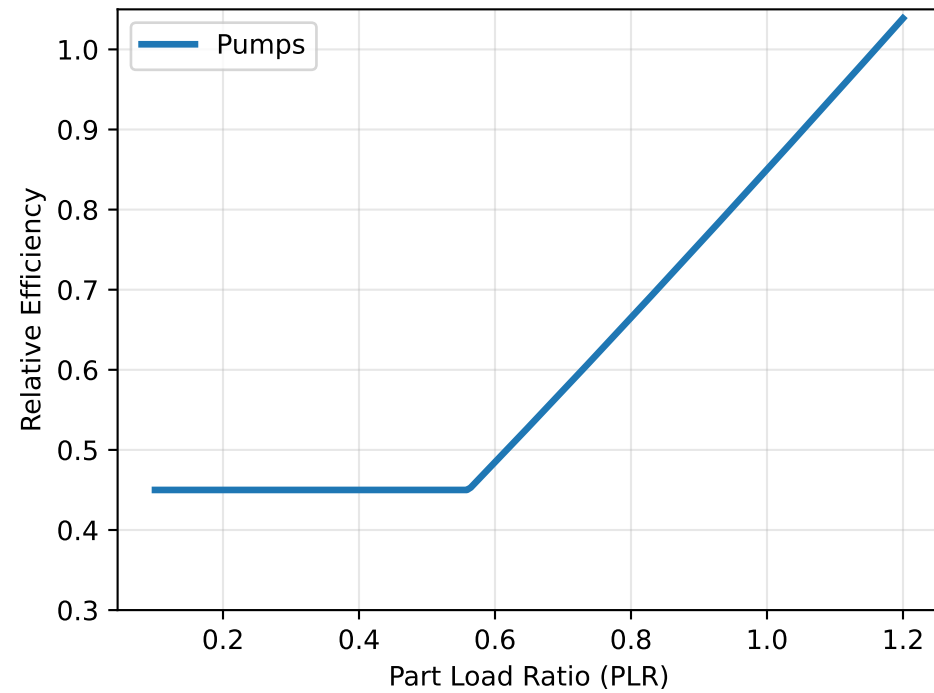
Part-Load Efficiency Curve: Boiler



Part-Load Efficiency Curve: Fans



Part-Load Efficiency Curve: Pumps



Part-Load Efficiency Curve: Lighting

